PS Analysis in Data Science: Problem Set 1

Problem 1. For $p \in [1, \infty)$, define $d_p : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$ by

$$d_p(x, y) = \left(\sum_{k=1}^{n} |x_k - y_k|^p \right)^{1/p}.$$

Each d_p defines a metric on $\mathbb{R}^n$.

(1) Implement the function $d_p(x, y)$ in Julia.

(2) Let $n = 2$. To illustrate the unit ball

$$B_1(0) = \{x \in \mathbb{R}^2 : d_p(x, 0) \leq 1\}$$

for $p = 1$, $p = 2$, and $p = 10$, create a grid of points in a square containing the ball and produce a scatter plot that includes only those grid points satisfying $d_p(x, 0) \leq 1$.

Describe how the shape of the ball changes as p increases.

(3) Show that

$$d_\infty(x, y) = \max_{k=1,\dots,n} |x_k - y_k|$$

defines a metric on $\mathbb{R}^n$.

Problem 2. Show that for every $p \in [1, \infty)$ and for every fixed $x, y \in \mathbb{R}^n$,

$$d_\infty(x, y) \leq d_p(x, y) \leq n^{1/p}\, d_\infty(x, y),$$

and deduce that $d_p(x, y) \to d_\infty(x, y)$ as $p \to \infty$.

Problem 3. Let (X, d_X) and (Y, d_Y) be metric spaces and let $f : X \to Y$ be a function. Show that f is continuous if and only if $f^{-1}(U)$ is open for all open sets $U \subseteq Y$.

(Hint: write the ε-δ definition of continuity in terms of balls)

Problem 4. Let (X, d_X) be a metric space and fix some $y_0 \in X$. Consider the mapping $f : X \to \mathbb{R}$,

$$f(x) = d_X(x, y_0)$$

and the balls for $\varepsilon > 0$ and $x \in X$,

$$B_\varepsilon(x) = \{y \in X : d_X(x, y) \leq \varepsilon\},$$
$$\mathring{B}_\varepsilon(x) = \{y \in X : d_X(x, y) < \varepsilon\}.$$

(1) Prove that f is continuous.

(2) Prove that $B_\varepsilon(x)$ is closed.

(3) Prove that $\mathring{B}_\varepsilon(x)$ is open.